

to find the set of values of the choice variables that will yield the desired extremum of the objective function.

17.4.1 Equality Constrained Optimisation

We have been given an objective function $f: R^n \rightarrow R$ to maximise, subject to k constraints. That is, there are k functions, $g^1: R^n \rightarrow R, g^2: R^n \rightarrow R, \dots, g^k: R^n \rightarrow R$, and we wish to

Maximise $f(x)$ over all $x \in R^n$ such that $g^1(x)=0, \dots, g^k(x)=0$.

More compactly, collect the constraint functions (looking them as component functions) into one function $g: R^n \rightarrow R$, where $g(x) = (g^1(x), \dots, g^k(x))$. Then what we want is to

Maximise $f(x)$ over all $x \in R^n$ such that $g(x)_{1 \times k} = \theta_{1 \times k}$.

The theorem of Lagrange provides the necessary conditions for a local optimum x^* . By local we mean that the value $f(x^*)$ is a maximum or minimum compared to other values $f(x)$ for all x contained in some open set U containing x^* such that x satisfies the k constraints. Thus, the problem considered is to provide necessary conditions for a maximum or a minimum of $f(x)$ over all $x \in S$, where $S = U \cap \{x \in R^n \mid g(x) = \theta\}$, for some open set U .

The following example illustrates the principle of ‘no arbitrage’ underlying a maximum. A more general illustration, with more than one constraint, requires a little bit of the machinery of linear inequalities for the illustration of which is out of the scope of this unit. However, such inequalities go a long way in optimisation theories of asset pricing, Nash equilibrium etc.

Example 17.7: A household has a utility function $U = U(x_1, x_2)$ with budget constraint $P_1x_1 + P_2x_2 = I$. Suppose x^* solves $\text{Max } U(x_1, x_2)$ subject to (s.t.), $I - P_1x_1 - P_2x_2 = 0$ and suppose $x^* > 0$. Then reallocating a small amount of income from one good to the other does not increase utility. Say, income $dI > 0$ is shifted from good 2 to good 1. So $dx_1 = (dI / P_1) > 0$ and $dx_2 = -(dI / P_2) < 0$. Note that this reallocation satisfies the budget constraint, since

$p_1(x_1 + dx_1) + p_2(x_2 + dx_2) = I$. The change in utility is $dU = U_1dx_1 + U_2dx_2$, or $dU = [(U_1 / p_1) - (U_2 / p_2)]dI \leq 0$, since the change in utility cannot be positive at a maximum. Therefore,

$$(U_1 / p_1) - (U_2 / p_2) \leq 0 \quad \dots\dots\dots(17.5)$$

Similarly, $dI > 0$ shifted from good 1 to good 2 does not increase utility, so that $[-(U_1 / p_1) + (U_2 / p_2)]dI \leq 0$, or $[-(U_1 / p_1) + (U_2 / p_2)] \leq 0 \quad \dots\dots\dots(17.6)$

Equations (17.5) and (17.6) imply

$$(U_1(x^*) / P_1) = (U_2(x^*) / P_2) = \lambda^* \quad \dots\dots\dots(17.7)$$

That is, the marginal utility of the last bit of income equals $(U_1(x^*)/P_1) = (U_2(x^*)/P_2)$ at the optimum. Also, equation 17.7 implies $U_1(x^*) = \lambda^* P_1$, $U_2(x^*) = \lambda^* P_2$. Along with $P_1 x_1^* + P_2 x_2^* = I$, these are the first order conditions of the Lagrangian function

$$\text{Max } L(x, \lambda) = U(x_1, x_2) + \lambda[I - P_1 x_1 - P_2 x_2]$$

Let us take a general case. Suppose $F: R^n \rightarrow R$ and $G: R^n \rightarrow R$, and suppose x^* solves $\text{Max } F(x)$ s.t. $c - G(x) = 0$.

Contemplate a change dx in x^* that respects the constraint $G(x) = c$.

That is, $dG = G_1 dx_1 + G_2 dx_2 = 0$.

Therefore, $G_1 dx_1 = -G_2 dx_2 = dc$ and $dx_1 = (dc/G_1)$, $dx_2 = (dc/G_2)$. If $dc > 0$, then change dx implies $dx_1 > 0$, $dx_2 < 0$. F does not increase at the maximum, x^* . So,

$$dF = F_1 dx_1 + F_2 dx_2 \leq 0,$$

$$\text{or, } [(F_1 / G_1) - (F_2 / G_2)]dc \leq 0.$$

Therefore, in the general case,

$$(F_1(x^*) / G_1(x^*)) = (F_2(x^*) / G_2(x^*)) = \lambda^* \quad \dots\dots\dots (17.8)$$

Caveat: We have assumed that $G_1(x^*)$ and $G_2(x^*)$ are not both zero at x^* . This is called the constraint qualification. Again, note that equation (17.8) can be taken as the first order condition (FOC) of the problem

$$\text{Max } L(x, \lambda) = F(x) + \lambda [c - G(x)].$$

Let's go back to the utility example. At the optimum (x^*, λ^*) , suppose you increase income by ΔI . Buying more x_1 implies utility increases by $(U_1(x^*)/p_1) \Delta I$, approximately. Buying more x_2 implies utility increases by $(U_2(x^*)/p_2) \Delta I$. At the optimum, $(U_1(x^*)/p_1) = (U_2(x^*)/p_2) = \lambda^*$. So in either case, utility increases by $\lambda^* \Delta I$, where λ^* gives the change in the objective function (here, the objective is utility), at an optimum that results from relaxing the constraint a little bit.

The interpretation is the same in the more general case: If $G(x) = c$, and c is increased by Δc . Suppose x_1 alone is then increased,

$$\text{so, } \Delta G = g_1 dx_1 = \Delta c,$$

$$\text{or, } dx_1 = (\Delta c / G_1).$$

That means, at x^* , F increases by $dF = F_1 dx_1 = (F_1(x^*)/G_1(x^*))\Delta c = \lambda^* \Delta c$. If instead x_2 is changed, F increases by $dF = F_2 dx_2 = (F_2(x^*)/G_2(x^*))\Delta c = \lambda^* \Delta c$.

Now we discuss the Lagrange theorem in detail. The set up is the following:

$f: R^n \rightarrow R$ is the objective function, $g^i: R^n \rightarrow R$, $i = 1, \dots, n$, $k, k < n$ are the constraint functions.

Let $g: R^n \rightarrow R^k$ be the function given by $g(x) = (g^1(x), \dots, g^k(x))$.
 $Df(x) = (\partial f(x) / \partial x_1, \dots, \partial f(x) / \partial x_n)$.

$$Dg(x) = \begin{pmatrix} (\partial g^1(x)/\partial x_1) & \dots & (\partial g^1(x)/\partial x_n) \\ \vdots & \ddots & \vdots \\ (\partial g^k(x)/\partial x_1) & \dots & (\partial g^k(x)/\partial x_n) \end{pmatrix} = \begin{pmatrix} Dg^1(x) \\ \vdots \\ Dg^k(x) \end{pmatrix}$$

Thus, $Dg(x)$ is a $k \times n$ matrix.

The theorem below provides a necessary condition for a local maximum or local minimum. Note that x^* is a local maximum (respectively, minimum) of f on the constraint set $\{x \in R^n \mid g^i(x) = 0, i = 1, \dots, k\}$ if $f(x^*) \geq f(x)$ (respectively for $\leq f(x)$) for all $x \in U$ for some open set U containing x^* , s.t. $g^i(x) = 0, i = 1, \dots, k$. Thus x^* is a maximum on the set $S = U \cap \{x \in R^n \mid g^i(x) = 0, i = 1, \dots, k\}$.

Theorem of Lagrange:

Let $f: R^n \rightarrow R$ and $g^1: R^n \rightarrow R, g^2: R^n \rightarrow R, \dots, g^k: R^n \rightarrow R$, $k < n$ be once continuously differentiable functions. Suppose x^* is a maximum or a minimum of f on the set $S = U \cap \{x \in R^n \mid g^i(x) = 0, i = 1, \dots, k\}$, for some open set $U \subseteq R^n$.

Then there exists real numbers $\mu^*, \lambda_1^*, \dots, \lambda_k^*$, not all zero, such that

$$\mu Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg^i(x^*) = \theta_{1 \times n}.$$

Moreover, if $\text{rank}(Dg(x^*)) = k$, then we may put $\mu = 1$.

Notes:

- 1) The condition $\text{rank}(Dg(x^*)) = k$ is called the constraint qualification. The first part of the theorem says that at a local maximum or minimum, under the assumption of continuous differentiability of f and $g^i, i = 1, \dots, k$, the vectors $Df(x^*), Dg^1(x^*), \dots, Dg^k(x^*)$ are linearly dependent. The constraint qualification (CQ) basically assumes that the vectors $Dg^1(x^*), \dots, Dg^k(x^*)$ are linearly independent. In that case, $\mu Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg^i(x^*) = \theta_{1 \times n}$ implies that μ cannot equal to zero, for then $\sum_{i=1}^k \lambda_i^* Dg^i(x^*) = \theta_{1 \times n}$, which along with linear independence implies $\lambda_i^* = 0, i = 1, \dots, k$. This cannot be. So, if the CQ holds, then $\mu \neq 0$, and we can divide through by μ .
- 2) In most applications, the CQ holds. We usually check first whether it holds, and then proceed. Suppose it does hold. Note that $\mu Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg^i(x^*) = \theta_{1 \times n}$ subsumes the following n equations:

$$(\partial f(x^*)/\partial x_j) + \sum_{i=1}^k \lambda_i^* (\partial g^i(x^*)/\partial x_j) = 0, j = 1, \dots, n.$$

Note also that this leads to the 'usual' procedure for finding equality constrained maximum or minimum, by setting up a Lagrangian function:

$$L(x, \lambda) = f(x) + \sum_{i=1}^k \lambda_i g^i(x),$$

and solving the FOC

$$(\partial L(x, \lambda) / \partial x_j) = (\partial f(x) / \partial x_j) + \sum_{i=1}^k \lambda_i (\partial g^i(x) / \partial x_j) = 0, j = 1, \dots, n$$

$$(\partial L(x, \lambda) / \partial \lambda_i) = g^i(x) = 0, i = 1, \dots, k \text{ which is } (n+k) \text{ equations in } (n+k)$$

variables $x_1, \dots, x_n, \lambda_1, \dots, \lambda_k$.

Second order condition: These conditions are characterised by definiteness or semi-definiteness of the Hessian of the Lagrangian function, which is the appropriate function to look at in this constrained optimisation problem. In addition, we don't have to check the appropriate inequality for the quadratic form for all x ; only those x are relevant that satisfy the constraints. Second order conditions in general capture the change of the curvature of the objective function around the local maximum or minimum, i.e., how the graph curves as we move from x^* to a nearby x . In constrained optimisation, we cannot move from x^* to any arbitrary x nearby; the move must be to an x , which satisfies the constraints. That is, such a move must leave all $g^i(x)$ at 0. In other words, $dg^i(x) = Dg^i(x).dx = 0$, where dx is a vector $x - x^*$ that must be orthogonal to $Dg^i(x)$. Thus, it suffices to evaluate the appropriate quadratic form at all vectors x that are orthogonal to all the gradients of the constraint functions.

$$\text{Since } L(x, \lambda) = f(x) + \sum_{i=1}^k \lambda_i g^i(x),$$

$$D^2 L(x, \lambda)_{n \times n} = D^2 f(x)_{n \times n} + \sum_{i=1}^k \lambda_i D^2 g^i(x)_{n \times n},$$

$$\text{where } D^2 f(x) = \begin{pmatrix} f_{11}(x) & \dots & f_{1n}(x) \\ \vdots & \ddots & \vdots \\ f_{n1}(x) & \dots & f_{nn}(x) \end{pmatrix} \text{ and } D^2 g^i(x) = \begin{pmatrix} g^i_{11}(x) & \dots & g^i_{1n}(x) \\ \vdots & \ddots & \vdots \\ g^i_{n1}(x) & \dots & g^i_{nn}(x) \end{pmatrix}$$

$$\text{So, } D^2 L(x, \lambda)_{n \times n} = \begin{pmatrix} f_{11}(x) + \sum_{i=1}^k \lambda_i g^i_{11}(x) & \dots & f_{1n}(x) + \sum_{i=1}^k \lambda_i g^i_{1n}(x) \\ \vdots & \ddots & \vdots \\ f_{n1}(x) + \sum_{i=1}^k \lambda_i g^i_{n1}(x) & \dots & f_{nn}(x) + \sum_{i=1}^k \lambda_i g^i_{nn}(x) \end{pmatrix} \text{ is the}$$

second order derivative of L with respect to the x variables. Note, that $D^2 L(x, \lambda)$ is symmetric so we may work with this quadratic form.

At a given $x^* \in R^n$,

$$Dg(x^*)_{k \times n} = \begin{pmatrix} Dg^1(x^*) \\ \vdots \\ Dg^k(x^*) \end{pmatrix}. \text{ So the set of all vectors } x \text{ that are orthogonal to all the}$$

gradient vectors of the constraint functions at x^* is the Null Space of $Dg(x^*)$, $N(Dg(x^*)) = \{x \in R^n \mid Dg(x^*)x = \theta_{k \times 1}\}$.

Theorem of Second Order Optimum:

Suppose there exists $(x^*_{n \times 1}, \lambda^*_{k \times 1})$ such that $\text{Rank}(Dg(x^*)) = k$

$$\text{and } Df(x^*) + \sum_{i=1}^k \lambda_i Dg^i(x^*) = \theta.$$

- i) A necessary condition: If f has a local maximum (respectively, local minimum) on S at point x^* , then $x^T D^2 L(x^*, \lambda^*) x \leq 0$, (respectively ≥ 0) for all $x \in N(Dg(x^*))$.
- ii) A sufficient condition: If $x^T D^2 L(x^*, \lambda^*) x < 0$, (respectively for > 0), for all $x \in N(Dg(x^*))$, $x \neq 0$, then x^* is a strict local maximum (respectively for strict local minimum) of f on S .

Note: The theorem means that for maximum, the second order condition should be that the bordered Hessian matrix of the problem is negative semi-definite (necessary) or negative definite (sufficient) given the FOC. For minimum, the second order condition should be positive semi-definite (necessary) or positive definite (sufficient given the FOC). Here the bordered Hessian is the matrix of second derivatives of $L(x, \lambda)$ with respect to x and λ .

Check Your Progress 2

- 1) Let $U(x, y) = x^2 y$ and $p_x=2$, $p_y=1$. Suppose, $I=200$. Find out x^* and y^* , if $I=p_x x + p_y y$ holds.

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- 2) Let $U(x, y) = x^2 + y$ and $p_x=2$, $p_y=1$. Suppose, $I=200$. Find out x^* and y^* , if $I=p_x x + p_y y$ holds.

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17.4.2 Inequality Constrained Optimisation

The problem is to find the maximum or the minimum of $f: R^n \rightarrow R$ on the set $\{x \in R^n \mid g^i(x) \geq 0, i=1, \dots, k\}$, where $g^i: R^n \rightarrow R$ are the k constraint functions. At the optimum, the constraints are now allowed to be binding (or tight or effective), i.e., $g^i(x) = 0$, as before, or slack (or non-binding), i.e., $g^i(x) > 0$.

Example 17.8: Max $U(x_1, x_2)$ s.t. $x_1 \geq 0$, $x_2 \geq 0$, $I - p_1 x_1 - p_2 x_2$. If we do not know whether $x_i = 0$, for some i , at the utility maximum, or whether $x_i > 0$, then clearly, we cannot use the theorem of Lagrange. Similarly, if there is a bliss point, then we do not know in advance whether there at the budget constrained optimum the budget constraint is binding or slack. Again, we cannot then use the theorem of Lagrange, to use which we need to be assured that the constraint is binding.

Note the general nature of a constraint of the form $g^i(x) \geq 0$. If we have a constraint $h(x) \leq 0$, this is equivalent to $-h(x) \geq 0$. And something like $h(x) \leq c$ is equivalent to $c - h(x) \geq 0$.

We use Kuhn-Tucker Theory to address optimisation problems with inequality constraints. The main result is a first order necessary condition that is slightly different from that of the theorem of Lagrange; the main difference is that the first order condition $g^i(x) = 0$, $i = 1, \dots, k$ in the theorem of Lagrange are replaced by the conditions $\lambda_i g^i(x) = 0$, $i = 1, \dots, k$ in Kuhn-Tucker theory.

17.5 KUHN-TUCKER THEORY

Recall that the problem is to find the maximum or the minimum of $f: R^n \rightarrow R$ on the set $\{x \in R^n \mid g^i(x) \geq 0, i = 1, \dots, k\}$, where $g^i: R^n \rightarrow R$ are the k constraint functions. The main theorem deals with local maxima and minima, though.

Suppose that l of the k constraints bind at the optimum x^* . Denote the corresponding constraint functions as $(g^i)_{i \in \Phi}$, where Φ is the set of indexes of the binding constraints. Let $g_\Phi: R^n \rightarrow R^l$ be the function whose components are the constraint functions of the binding constraints. That is, $g_\Phi(x) = (g^i(x))_{i \in \Phi}$.

$$Dg_\Phi(x) = \begin{pmatrix} Dg^i(x) \\ \vdots \\ Dg^m(x) \end{pmatrix}, \text{ where } i, \dots, m \text{ are the indexes of the binding constraints.}$$

So, $Dg_\Phi(x)$ is a $l \times n$ matrix.

We now state FOC for the problem. The theorem below is a consolidation of the Fritz-John and the Kuhn-Tucker theorems.

17.5.1 The Kuhn-Tucker (KT) Theorem

Let $f: R^n \rightarrow R$, and $g^i: R^n \rightarrow R$, $i = 1, \dots, k$

be once continuously differentiable functions. Suppose x^* is a maximum of f on the set

$$S = U \cap \{x \in R^n \mid g^i(x) \geq 0, i = 1, \dots, k\}, \text{ for some open set } U \subseteq R^n.$$

Then there exist real numbers $\mu, \lambda_1^*, \dots, \lambda_k^*$, not all zero such that $\mu Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg^i(x^*) = \theta_{1 \times n}$.

Moreover, if $g^i(x^*) > 0$ for some i , then $\lambda_i^* = 0$. If in addition, $\text{rank } Dg_\Phi(x^*) = l$, then we may take μ to be equal to 1. Furthermore, $\lambda_i^* \geq 0$, $i = 1, \dots, k$ and $\lambda_i^* > 0$ for some l implies $g^i(x^*) = 0$.

Suppose the constraint qualification, $\text{Rank } Dg_\Phi(x^*) = l$, is met at the optimum. Then the KT equations are the following $(n+k)$ equations in the $n+k$ variables $x_1, \dots, x_n, \lambda_1, \dots, \lambda_k$.

$$\lambda_i g^i(x^*) = 0, i = 1, \dots, k, \lambda_i^* \geq 0, g^i(x^*) \geq 0 \text{ with complementary slackness} \quad \dots (17.9)$$

$$Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg^i(x^*) = \theta \quad \dots (17.10)$$

If x^* is a local minimum of f on S , then $-f(x^*)$ attains a local maximum value. Thus, for minimisation, while equation (17.9) stays the same, equation (17.10) changes to

$$-Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg^i(x^*) = \theta \quad \dots (17.10a)$$

Equation (17.9) and (17.10) are known as the Kuhn-Tucker conditions. Note finally that since the conditions of KT theorem are not sufficient conditions for local optima, there may be points that satisfy equations (17.9) and (17.10) or (17.10a) without being local optima. This method does not require checking for concavity of objective functions and constraints, and does not require checking any second order condition.

Next, we give one useful theorem that gives some link between local and global maxima, where the crucial assumption is that the objective function is concave.

Theorem: Suppose $S \subseteq R^n$ is a convex set, and $f: S \rightarrow R$ is concave. Then

- 1) Any local maximum of f is a global maximum of f .
- 2) The set of maximisers of f on S , $\arg \max \{f(x) | x \in S\}$ is either empty or convex.

Check Your Progress 3

- 1) Consider the following utility maximisation problem, in which the utility function of the consumer is given by $-U(x, y) = x^{0.5}y^{0.5}$. The consumer income is $I = 100$ and it is given to him that $p_x = 2$ and $p_y = 3$. Find the utility maximising x and y when the budget constraint of the consumer may or may not bind (Apply the Kuhn-Tucker method).

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- 2) Maximize $U(x, y) = (x - 3)(y - 2)$ subject to the following constraints:

- i) $x \geq 0$.
- ii) $y \geq 0$.
- iii) $5x + 3y \leq 50$.

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17.6 SIMPLE ECONOMIC PROBLEMS: CONSTRAINED OPTIMISATION

In this section, we present two simple utility maximisation problems.

Example 17.9: In this problem our objective is to

maximise $U(x_1, x_2) = x_1 + x_2$,

over the set $\{x = (x_1, x_2) \in \mathbb{R}^2 \mid x_1 \geq 0, x_2 \geq 0, I - p_1x_1 - p_2x_2 = 0\}$,

where $I > 0$, $p_1 > 0$ and $p_2 > 0$ are given.

So, there are three inequality constraints:

$$g^1(x_1, x_2) = x_1 \geq 0, \quad g^2(x_1, x_2) = x_2 \geq 0 \quad \text{and} \quad g^3(x_1, x_2) = I - p_1x_1 - p_2x_2 \geq 0$$

At the maximum x^* , any combination of these three could bind; so there are eight possibilities. However, since U is strictly increasing, the budget constraint binds at the maximum, therefore $g^3(x^*) = 0$. Moreover, $g^1(x^*) = g^2(x^*) = 0$ is not possible since consuming 0 of both goods gives utility equal to 0, which is clearly not a maximum.

Therefore, we have to check just three possibilities out of the eight.

Case (1) $g^1(x^*) > 0, g^2(x^*) > 0, g^3(x^*) = 0$

Case (2) $g^1(x^*) = 0, g^2(x^*) > 0, g^3(x^*) = 0$

Case (3) $g^1(x^*) > 0, g^2(x^*) = 0, g^3(x^*) = 0$

Before using the Kuhn Tucker conditions, we verify that

- i) A global max exists (here, because the utility function is continuous and the budget set is compact).
- ii) The CQ holds at all three relevant combinations of binding constraints described above.

Indeed, for Case (1), $Dg_\Phi(x) = Dg^3(x) = (-p_1, -p_2)$, so $\text{Rank}[Dg^3(x)] = 1$, so CQ holds. For Case (2), $Dg_\Phi(x) = \begin{pmatrix} Dg^1(x) \\ Dg^2(x) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -p_1 & -p_2 \end{pmatrix}$, so $\text{Rank}[Dg_\Phi(x)] =$

2. For Case (3), $Dg_\Phi(x) = \begin{pmatrix} Dg^2(x) \\ Dg^3(x) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -p_1 & -p_2 \end{pmatrix}$, so $\text{Rank}[Dg_\Phi(x)] = 2$.

Thus, for the maximum, x^* , there exists a λ^* such that (x^*, λ^*) will be a solution to the KT FOCs. Of course, there could be other (x, λ) 's that are solutions as well, but a simple comparison of $U(x)$ for all candidate solutions will isolate for us the maximum.

$$L(x, \lambda) = x_1 + x_2 + \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 (I - p_1 x_1 - p_2 x_2)$$

The KT conditions are –

$$\lambda_1 (\partial L / \partial \lambda_1) = \lambda_1 x_1 = 0, \lambda_1 \geq 0, x_1 \geq 0, \text{ with CS} \quad (17.11)$$

... (1)

$$\lambda_2(\partial L/\partial \lambda_2) = \lambda_2 x_2 = 0, \lambda_2 \geq 0, x_2 \geq 0, \text{ with CS} \quad \dots (17.12)$$

$$\lambda_3(\partial L/\partial \lambda_3) = \lambda_3(I - p_1 x_1 - p_2 x_2) = 0, \lambda_3 \geq 0, (I - p_1 x_1 - p_2 x_2) \geq 0,$$

$$\text{with CS} \quad \dots (17.13)$$

$$(\partial L/\partial x_1) = 1 + \lambda_1 - \lambda_3 p_1 = 0 \quad \dots (17.14)$$

$$(\partial L/\partial x_2) = 1 + \lambda_2 - \lambda_3 p_2 = 0 \quad \dots (17.15)$$

Since we don't know which of these three cases select the constraints that bind at the maximum, we must try all three.

Case 1: Since $x_1 > 0$, $x_2 > 0$, (17.11) and (17.12) imply $\lambda_1 = \lambda_2 = 0$. Plugging these in equation (17.14) and (17.15), we have $1 = \lambda_3^* p_1 = \lambda_3^* p_2$. Since utility is strictly increasing, relaxing the budget constraint will increase utility. So, the marginal utility of income is $\lambda_3^* > 0$. Thus $\lambda_3^* p_1 = \lambda_3^* p_2$ implies $p_1 = p_2$. From equation (17.14) it can also be seen that $\lambda_3^* = 1/p_1 > 0$.

Therefore, if at a local maximum both x_1 and x_2 are strictly positive, then it must be that their prices are equal. All (x_1, x_2) that solve equation (17.13) are solutions. The utility in any such case equals $x_1 + ((I - p_1 x_1)/p_2) = I/p$, where $p = p_1 = p_2$. Note that in this case, $(U_1/p_1) = (U_2/p_2) = 1/p$.

Case 2: $x_1 = 0$ implies, from equation (17.13), that $x_2 = (I/p_2)$. Since this is greater than 0, equation (17.12) implies $\lambda_2 = 0$. Hence from equation (5), $\lambda_3^* p_2 = 1$. Since $\lambda_1^* \geq 0$, equation (17.14) and (17.15) imply $\lambda_3^* p_1 = 1 + \lambda_1 \geq 1 = \lambda_3^* p_2$. Moreover, since $\lambda_3^* > 0$, this implies $p_1 \geq p_2$.

That is, if it is the case that at the maximum, $x_1^* = 0$, $x_2^* > 0$, then it must be that $p_1 \geq p_2$. Note that in this case,

$$(U_2/p_2) = (1/p_2) \geq (U_1/p_1) = (1/p_1).$$

For completeness sake, equation (17.15) implies $\lambda_3^* = (1/p_2)$. So from equation (17.14), $\lambda_1^* = (p_1/p_2) - 1$. So the unique critical point of $L(x, \lambda)$ is

$$(x^*, \lambda^*) = (x_1^*, x_2^*, \lambda_1^*, \lambda_2^*, \lambda_3^*) = (0, (I/p_2), (p_1/p_2) - 1, 0, (1/p_2)).$$

Case 3: This case is similar, and we get that $x_2^* = 0$, $x_1^* > 0$ occurs only if $p_1 \leq p_2$. We have $(x^*, \lambda^*) = ((I/p_2), 0, 0, (p_2/p_1) - 1, 1/p_1)$.

We see that which cases apply depends upon the price ratio (p_2/p_1) . If $(p_2/p_1) = 1$, then all these three cases are relevant, and all $(x_1, x_2) \in R_+^2$ such that the budget constraint binds are utility maxima. But if $(p_2/p_1) > 1$, then only Case (2) is applies, and the solution to the KT conditions in that case is the utility maximum. Similarly, if $(p_2/p_1) < 1$, only Case (3) applies.

Example 17.10: Max $U(x_1, x_2) = (x_1/(1+x_1)) + (x_2/(1+x_1))$, subject to $x_1 \geq 0$, $x_2 \geq 0$, $p_1 x_1 + p_2 x_2 \leq I$.

Check that the indifference curves are downward sloping, convex and that they cut the axes. This last condition is due to the additive form of the utility function,

and may result in 0 consumption of one of the goods at the utility maximum. Exactly as in the previous Example, we are assured that a global max exists, that the CQ is met at the optimum, and that there are only three relevant cases of binding constraints to check.

The Kuhn-Tucker conditions are:

$$\lambda_1(\partial L / \partial \lambda_1) = \lambda_1 x_1 = 0, \lambda_1 \geq 0, x_1 \geq 0, \text{ with CS} \quad \dots(17.16)$$

$$\lambda_2(\partial L / \partial \lambda_2) = \lambda_2 x_2 = 0, \lambda_2 \geq 0, x_2 \geq 0, \text{ with CS} \quad \dots (17.17)$$

$$\lambda_3(\partial L / \partial \lambda_3) = \lambda_3(I - p_1 x_1 - p_2 x_2) = 0, \lambda_3 \geq 0, (I - p_1 x_1 - p_2 x_2) \geq 0, \text{ with CS} \quad \dots(17.18)$$

$$(\partial L / \partial x_1) = \left(\frac{1}{(1+x_1)^2} \right) + \lambda_1 - \lambda_3 p_1 = 0 \quad \dots(17.19)$$

$$(\partial L / \partial x_2) = \left(\frac{1}{(1+x_2)^2} \right) + \lambda_2 - \lambda_3 p_2 = 0 \quad \dots(17.20)$$

Case 1: $x_1 > 0, x_2 > 0$ implies $\lambda_1 = \lambda_2 = 0$. Equation (17.19) implies $\lambda_3 > 0$, so that

$$\text{equation (17.19) and (17.20) give } \left(\frac{(1+x_1)}{(1+x_2)} \right) = \left(\frac{p_1}{p_2} \right)^{\frac{1}{2}}.$$

Using equation (17.18), which gives $x_2 = \left(\frac{(I - p_1 x_1)}{p_2} \right)$, above, we get

$$\left(\frac{(p_2 + I - p_1 x_1)}{(p_2 (1+x_1))} \right) = \left(\frac{p_1}{p_2} \right)^{\frac{1}{2}}$$

Therefore, simple consumptions yield

$$x_1 = \frac{\left(I + p_2 - (p_1 p_2)^{\frac{1}{2}} \right)}{\left(p_1 + (p_1 p_2)^{\frac{1}{2}} \right)}$$

$$x_2 = \frac{\left(I + p_1 - (p_1 p_2)^{\frac{1}{2}} \right)}{\left(p_2 + (p_1 p_2)^{\frac{1}{2}} \right)}$$

$$\lambda_3 = \frac{1}{p_1 (1+x_1)^2} = \frac{1}{p_2 (1+x_2)^2}$$

$x_1 > 0, x_2 > 0$ implies $I > (p_1 p_2)^{\frac{1}{2}} - p_1, I > (p_1 p_2)^{\frac{1}{2}} - p_2$. If either of these fails, then we are not in the regime of Case (1).